Team Problems

Note: In 2014, the Theme Round was removed and the Team Round was expanded to 15 questions and 45 minutes.

1. There are two times between noon and 1 pm where the hour and minute hands of a clock are perpendicular. Compute the number of minutes between these two times.

2. The three distinct integer roots of $x^3 + qx^2 - 2qx - 8 = 0$ form an arithmetic progression. Compute q.

3. Suppose that n leaves a remainder of 24 when divided by 77. If n leaves a remainder of A when divided by 7 and a remainder of B when divided by 11, compute $A + B$.

4. Compute the length of the interval of values x for which $\frac{1}{x+\sqrt{x}} + \frac{1}{x-\sqrt{x}} \geq 1$.

5. Compute the number of positive integer factors of $N = 1{,}004{,}006{,}004{,}001$.

6. For a positive integer n, let $p(n)$ be the product of the digits of n. Compute the number of positive integers less than 1000 for which $p(p(n)) = 6$.

7. The continued fraction $1+\cfrac{1}{3+\cfrac{1}{1+\cfrac{1}{3+\cfrac{1}{\ddots}}}} = \frac{A+\sqrt{B}}{C}$ for integers A, B, and C. Compute the minimum value of $A + B + C$.

8. Compute the number of ways to color the cells of a 3×3 grid red, green, or blue such that each color appears in at least two cells and no cells that share an edge have the same color. Note that reflections and rotations of a grid are considered distinct.

9. A quadrilateral $MATH$ is inscribed in a circle of radius 10. $MA = AT = 12$ and $MH = TH = 16$. Compute the radius of the circle inscribed within $MATH$.

10. Let the legs of a right triangle have lengths x and y. If $x + y = 98$ and $x^3 + y^3 = 517832$, compute the area of the triangle.

11. The expression $n^2 + n - 9$ is a multiple of 101 for two positive integers less than 100. One of these integers is 10. Compute the other one.

12. The graph of the line $y = kx$ intersects the graph of $y = |x - 4|$ at A and B, where A is to the left of B. Let O be the origin. If $OA = AB$, compute k.

13. Two circles pass through the points $(9, 0)$ and $(8, 7)$ and are tangent to the y-axis. Compute the sum of the radii of these two circles.

14. If $2\,(\underline{G}\,\underline{E}\,\underline{R}\,\underline{M}\,\underline{A}\,\underline{N}) = 5\,(\underline{M}\,\underline{A}\,\underline{N}\,\underline{G}\,\underline{E}\,\underline{R})$, and every letter corresponds to a different non-zero digit, compute the value of $\underline{E}\,\underline{N}\,\underline{G}\,\underline{R}\,\underline{A}\,\underline{M}$.

15. An *interior diagonal* of a polyhedron P is a line segment whose endpoints are vertices of P and which lies entirely in the interior of P (that is, does not intersect any faces of P), except for its endpoints. Find the maximum possible number of interior diagonals if P has 12 vertices.